1. Start cqlsh

cqlsh

```
docker exec -it 97caed39b67
# cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.10 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh>
```

2. Create Keyspace with Replication Factor = 3

```
CREATE KEYSPACE lab_ks
WITH replication = {
  'class': 'SimpleStrategy',
  'replication_factor': 3
};
```

```
cqlsh> CREATE KEYSPACE lab_ks
   ... WITH replication = {
   ...   'class': 'SimpleStrategy',
   ...   'replication_factor': 3
   ... };

Warnings :
Your replication factor 3 for keyspace lab_ks is higher than the number
of nodes 1

cqlsh>
```

USE lab_ks;

```
cqlsh> USE lab_ks;
cqlsh:lab_ks>
```

3. Create Table

```
CREATE TABLE users (
  id INT PRIMARY KEY,
  name TEXT,
  city TEXT
);
```

```
cqlsh> USE lab_ks;
cqlsh:lab_ks> CREATE TABLE users (
          ...   id INT PRIMARY KEY,
          ...   name TEXT,
          ...   city TEXT
          ... );
cqlsh:lab_ks>
```

4. Test Consistency Level: ONE

Set consistency to ONE

CONSISTENCY ONE;

```
          ... );
cqlsh:lab_ks> CONSISTENCY ONE;
Consistency level set to ONE.
```

Insert Data

INSERT INTO users (id, name, city) VALUES (1, 'Alice', 'Bangalore');

Read Data

SELECT * FROM users WHERE id = 1;

```
cqlsh:lab_ks> SELECT * FROM users WHERE id = 1;

 id | city      | name
----+-----------+-------
  1 | Bangalore | Alice

(1 rows)
```

Observation: Fast response, but may return stale data.

5. Test Consistency Level: QUORUM

Set consistency to QUORUM

CONSISTENCY QUORUM;

```
cqlsh:lab_ks> CONSISTENCY QUORUM;
Consistency level set to QUORUM.
cqlsh:lab_ks>
```

Insert Data

INSERT INTO users (id, name, city) VALUES (2, 'Bob', 'Mysore');

Read Data

SELECT * FROM users WHERE id = 2;

```
cqlsh:lab_ks> SELECT * FROM users WHERE id = 2;

 id | city   | name
----+--------+------
  2 | Mysore |  Bob

(1 rows)
```

Observation: Balanced consistency and latency.

6. Test Consistency Level: ALL

Set consistency to ALL

CONSISTENCY ALL;

```
cqlsh:lab_ks> CONSISTENCY ALL;
Consistency level set to ALL.
cqlsh:lab_ks>
```

7. Check Current Consistency Level

CONSISTENCY;

```
cqlsh:lab_ks> CONSISTENCY;
Current consistency level is ALL.
cqlsh:lab_ks>
```

8. Cleanup (Optional)

DROP KEYSPACE lab_ks;